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Gilberto Recupito

1 Personal Information

Full Name

Gilberto Recupito

Date of Birth

January 25th, 1997

Place of Birth

Salerno, Italy

Current Address

Via Roma 2 - 84084, Fisciano, Salerno, Italy

2 Current Position

Postdoctoral Researcher

With Prof. Coen De Roover at VUB Brussels University

Feb 2026 – current

Bruxelles, Belgium

3 Work Experience

Research Intern

Project collaboration with Prof. Foutse Khomh at École Polytechnique de Montréal

Mar 2025 – May 2025

Montréal, Canada

Guest Researcher

Project collaboration with Prof. Michael Felderer at German Aerospace center (DLR)

May 2024 – Jul 2024

Cologne, Germany

Research Assistant

CloudSea.AI, research group with Prof. Davide Taibi at Tampere University

Apr 2022 – Sep 2022

Tampere, Finland

4 Education

Ph.D. in Computer Science

2025

Excellent, Advisors: Profs. Fabio Palomba & Dario Di Nucci

Università degli studi di Salerno

Master's Degree (MSc) in Computer Science

2022

Magna cum laude, Advisor: Prof. Fabio Palomba

Università degli studi di Salerno

Bachelor's Degree (BSc) in Computer Science

2019

Advisor: Prof. Vittorio Scarano

Università degli studi di Salerno

5 Research Areas

My primary research area is **Software Engineering for Artificial Intelligence (SE4AI)**. My work advances the engineering of AI-intensive systems by:

- **AI Technical Debt:** Characterizing and mitigating sources of technical debt in AI-enabled systems, including system-level [1], data-centric [2], and ML-specific code quality issues [3], to improve reliability and maintainability.
 - **MLOps and Engineering of ML Pipelines:** Defining practices, features, and tools that support the development, automation, and operation of robust ML pipelines [4]. This includes empirical analysis of ML projects, enabling the characterization of *ML-Model Producers* and *ML-Model Consumers* [5].
 - **LLM4SE and Agentic AI:** Investigating how Large Language Models can support software engineering tasks, with a focus on the design of prompt patterns to manage complexity in LLM-generated code and support code quality evaluation [6].
 - **Empirical Software Engineering & Mining Software Repositories:** Contributing to empirical methods and MSR research, including educational work culminating in a book chapter to support instructors in teaching MSR techniques [7].
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6 Professional Activities

6.1 Professional Recognitions

Distinguished Reviewer Award

2026

FORGE 2026

Rio de Janeiro, Brazil

6.2 Organizing Activities

3rd Workshop on Software Quality Assurance for Artificial Intelligence (SQA4AI)	2026
<i>Organizer</i>	<i>Limassol, Cyprus</i>

6.3 Committee Activities

International Conference on Software Maintenance and Evolution (ICSME)	2026
<i>Tool Demo and Data Showcase Track Program Committee</i>	<i>Benevento, Italy</i>
<i>Industry Track Program Committee</i>	
<i>Replication and Negative Results Track Program Committee</i>	

International Conference on AI Engineering (CAIN)	2026
<i>Poster Track Program Committee</i>	<i>Rio De Janeiro, Brazil</i>

1st ECAI Workshop on MLOps (MLOps)	2025
<i>Session Chair</i>	<i>Bologna, Italy</i>

International Conference on AI Foundation Models and Software Engineering (FORGE)	2026
<i>Program Committee</i>	<i>Rio De Janeiro, Brazil</i>

Euromicro Conference on Software Engineering and Advanced Applications (SEAA)	2025
<i>Session Chair</i>	<i>Salerno, Italy</i>

International Conference on AI Foundation Models and Software Engineering (FORGE)	2025
<i>Program Committee</i>	<i>Ottawa, Canada</i>

International Conference on Software Analysis, Evolution and Reengineering (SANER)	2025
<i>Session Chair</i>	<i>Montréal, Canada</i>

2nd Workshop on Software Quality Assurance for Artificial Intelligence (SQA4AI)	2025
<i>Program Committee Member</i>	<i>Montréal, Canada</i>

International Conference on Automated Software Engineering (ASE)	2024
<i>Artifact Evaluation Track Program Committee</i>	<i>California, United States</i>

International Conference on AI Foundation Models and Software Engineering (FORGE) <i>Program Committee</i>	2024 <i>Lisbon, Portugal</i>
International Working Conference on Mining Software Repositories (MSR) <i>Junior Program Committee</i>	2024 <i>Lisbon, Portugal</i>
Seminar Series on Advanced Techniques & Tools for Software Evolution (SATToSE) <i>Social Media Chair</i>	2023 <i>Salerno, Italy</i>
International Conference on Software Engineering Advances (ICSEA) <i>Program Committee Member</i>	2023 <i>Valencia, Spain</i>
Annual Symposium for Computer Science <i>Organizing Committee Member</i>	2022 <i>Tampere, Finland</i>

6.4 Journal Services

Journal of Systems and Software <i>Guest Editor - Special Issue Software Quality Assurance for AI</i>	2025, current
Automated Software Engineering Journal <i>Reviewer</i>	2025
SoftwareX <i>Reviewer</i>	2024, 2025
Journal of Software: Evolution and Process <i>Reviewer</i>	2024
ACM Transactions on Software Engineering and Methodology <i>Reviewer</i>	2023-2025
Empirical Software Engineering <i>Reviewer</i>	2024, 2025
IET Software <i>Reviewer</i>	2025
ACM Conference On Computer-Supported Cooperative Work And Social Computing <i>Reviewer</i>	2023, 2024
Journal of Systems and Software <i>Reviewer</i>	2023, 2024
Science of Computer Programming <i>Reviewer</i>	2023
Journal of Decision Systems <i>Reviewer</i>	2022

6.5 Teaching Activities

During the three years of Ph.D., I provided support and served as guest lecturer for the following courses:

6.5.1 University of Salerno, Italy

Software Engineering (B.Sc.) <i>Lectures on LLM-enabled software development.</i>	2024, 2025 Prof. Di Nucci
Sustainable Software Engineering (M.Sc.) <i>Guest Lecture on Technical Debt.</i>	2025 Prof. Di Nucci
Software Dependability (M.Sc.) <i>Support to course projects.</i>	2023, 2024 Prof. Di Nucci
Software Engineering for Artificial Intelligence (M.Sc.) <i>Support to course projects and lectures on Transfer Learning, Transformers and MLOps.</i>	2023, 2024 Prof. Palomba
Software Engineering, Management and Evolution (M.Sc.) <i>Support to course projects and lectures on DevOps.</i>	2023, 2024 Prof. De Lucia
Software Engineering (B.Sc.) <i>Support to course projects and guest lectures on Maven, GitHub, and testing frameworks.</i>	2023, 2024 Prof. De Lucia

6.6 Advising Activities

In my career, I had the opportunity to advise Bachelors and Masters students in their career.

- MLOps: A systematic review of the literature on MLOps practices and properties. *Damiana Buono, Bachelor's Degree (2023).*
- Data Smells: towards the exploration and analysis of the impact on data quality. *Raimondo Rapacciuolo, Master's Degree (2023).*
- CAMILLE: A chatbot for ML-specific code Smells identification and refactoring. *Francesco Pinto, Bachelor's Degree (2024).*
- Analysis and Exploration of MLOps tools for monitoring data drift - *Ranjeet Kumar, Bachelor's Degree (2024).*
- Synthetic data smell injection: Effect and impact on quality in ML-enabled systems. *Davide La Gamba, Master's Degree (2024).*
- Machine Learning Detection of Data Smells. *Nicolò Gallotta, Bachelor's Degree (2024).*
- Multi-Agent System for information needs extraction of code review tasks. *Giulio Palladino, Bachelor's Degree (2024).*
- CRANE - Code Review AI Network Engine. *Luca Morelli, Master's Degree (2025).*

- Development and testing of CodeSmile: a plugin for the automatic analysis of Machine Learning Code Smells. *Daniele Pio Scaparra, Bachelor's Degree (2025).*
- Analysis of the Correlation between Data Smells, Performance, Fairness, and Sustainability in Machine Learning-Based Systems. *Domenico D'Antuono, Master's Degree (2025).*
- Analysis of the evolution of code smells in Machine Learning-Based Systems. *Simone Silvestri, Master's Degree (2025).*

6.7 Guest Lectures

Vrije Universiteit Brussel

2026

Guest Lecture on Large Language Models for Software Engineering

Brussel, Belgium

7 Conferences and School Participations

International Workshop of Software Quality Assurance for Artificial Intelligence (SQA4AI)

2024

Presentation of [8] and [6]

Salerno, Italy

International Conference on Evaluation and Assessment in Software Engineering (EASE)

2024

Attended

Salerno, Italy

3rd International Conference on AI Engineering (CAIN)

2024

Presentation of [2]

Salerno, Italy

Seminar Series on Advanced Techniques & Tools for Software Evolution (SATToSE)

2023

Speaker

Salerno, Italy

International School of Software Engineering (ISSSE)

2023

Attended

Salerno, Italy

48th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)

2022

Presentation of [4]

Gran Canaria, Spain

8 Additional Contributions

VLAIO Project: Cegeka-Next

2026

Actively contributed to the VLAIO Research Foundation project. Activity includes organization and participation in knowledge sharing workshops and in-practice experiment to support quality and productivity monitoring in agent-assisted development.

FWO Project: CodeGuard

2026

Actively contributed to the Flanders Research Foundation project. Activities include research and knowledge sharing to support LLM and agents secure code generation

PRIN Project: FRINGE

2023-2025

Actively contributed to the proposal of the winning PRIN project FRINGE awarded by NextGeneration EU and the MUR (Ministero dell'Università e della Ricerca).

PRIN Project: QUALAI

2022-2025

Actively contributed to the PRIN project QUALAI awarded by the MUR (Ministero dell'Università e della Ricerca).

9 Publications

- [1] G. Recupito, F. Pecorelli, G. Catolino, V. Lenarduzzi, D. Taibi, D. Di Nucci, and F. Palomba, “Technical debt in ai-enabled systems: On the prevalence, severity, impact, and management strategies for code and architecture,” *Journal of Systems and Software*, p. 112 151, 2024, ISSN: 0164-1212. DOI: <https://doi.org/10.1016/j.jss.2024.112151>.
- [2] G. Recupito, R. Rapacciuolo, D. Di Nucci, and F. Palomba, “Unmasking data secrets: An empirical investigation into data smells and their impact on data quality,” *3rd International Conference on AI Engineering – Software Engineering for AI*, 2024.
- [3] G. Recupito, G. Giordano, F. Ferrucci, D. Di Nucci, and F. Palomba, “When code smells meet ml: On the lifecycle of ml-specific code smells in ml-enabled systems,” in *Empirical Software Engineering (EMSE)*, 2025.
- [4] G. Recupito, F. Pecorelli, G. Catolino, S. Moreschini, D. Di Nucci, F. Palomba, and D. A. Tamburri, “A multivocal literature review of ml ops tools and features,” *Euromicro Conference on Software Engineering and Advanced Applications*, 2022. DOI: <http://dx.doi.org/10.13140/RG.2.2.10257.71526>.
- [5] V. De Martino, G. Recupito, G. Giordano, F. Ferrucci, D. Di Nucci, and F. Palomba, “Into the ml-universe: An improved classification and characterization of machine-learning projects,” *Journal of Systems and Software*, 2025.

- [6] A. Della Porta, G. Recupito, S. Lambiase, D. Di Nucci, and F. Palomba, “Unlocking code simplicity: The role of prompt patterns in managing llm code complexity,” in *International Workshop of Software Quality Assurance for Artificial Intelligence (SQA4AI)*, 2025.
- [7] Z. Codabux, F. Fard, R. Verdecchia, F. Palomba, D. Di Nucci, and G. Recupito, “Teaching mining software repositories,” in *Handbook on Teaching Empirical Software Engineering*. Springer, 2024.
- [8] G. Recupito, V. De Martino, D. Di Nucci, and F. Palomba, “A first look at the lifecycle of dl-specific self-admitted technical debt,” in *International Workshop of Software Quality Assurance for Artificial Intelligence (SQA4AI)*, 2025.
- [9] G. Recupito, G. Giordano, D. Di Nucci, and F. Palomba, “Detecting semantic data smells with bert: A transformer-based approach to data quality,” 2025.
- [10] F. Palomba, G. Voria, A. Parziale, V. Pentangelo, A. D. Porta, V. D. Martino, G. Recupito, and G. Giordano, “Teaching software engineering for artificial intelligence: An experience report,” in *Euromicro Conference on Software Engineering and Advanced Applications*, Springer, 2025, pp. 214–230.
- [11] D. La Gamba, G. Iuliano, G. Recupito, G. Giordano, F. Ferrucci, D. Di Nucci, and F. Palomba, “Toward a search-based approach to support the design of security tests for malicious network traffic,” in *Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering*, ser. EASE ’24, , Salerno, Italy, Association for Computing Machinery, 2024, pp. 624–628, ISBN: 9798400717017. DOI: 10.1145/3661167.3661288.
- [12] N. Novielli, R. Oliveto, F. Palomba, F. Calefato, G. Colavito, V. De Martino, A. Della Porta, G. Giordano, E. Guglielmi, F. Lanubile, *et al.*, “Qualai: Continuous quality improvement of ai-based systems,” in *RCIS Workshops*, 2024.
- [13] N. Novielli, R. Oliveto, F. Palomba, F. Calefato, G. Colavito, V. De Martino, A. Della Porta, G. Giordano, E. Guglielmi, F. Lanubile, *et al.*, “Continuous quality improvement of ai-based systems: The qualai project,” in *Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement*, 2024, pp. 603–607.
- [14] A. Della Porta, V. De Martino, G. Recupito, C. Iemmino, G. Catolino, D. Di Nucci, and F. Palomba, “Using large language models to support software engineering documentation in waterfall life cycles: Are we there yet?,” 2024.

In Fede: